Chen 2021 Prostate Cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Integration + continuity: iLISI rises from a sample-segregated baseline (median 1.0) and plateaus by θ=10 (median 3.6, 12 samples intermixed); 0 detached islands — one continuous manifold.

Gate 2 — module geography → GO (non-branching)

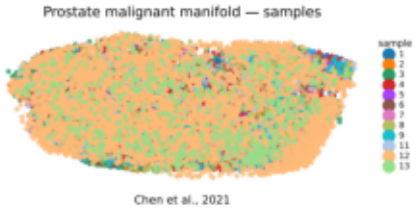
Shared Gast/pSom developmental gradient with Crest as an inverse terminal gradient (pSom↔Crest r=−0.76; Crest-high decile at pseudotime 0.92). No two competing arms.

Gate 3 — hub cross-repression → NA

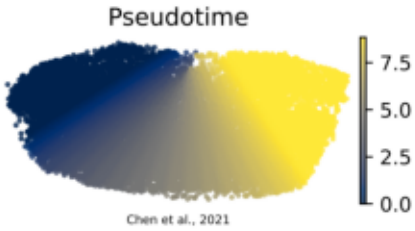
No bifurcation: a PPT tree over-forks a continuous cloud — all tips share one profile (low Gast, moderate pSom, low Crest), no opposing Gast-high vs pSom-high arms. Curve fit: 2 tips, fold-free.

STEP FIGURES

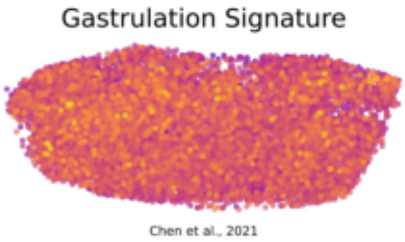
Manifold — samples



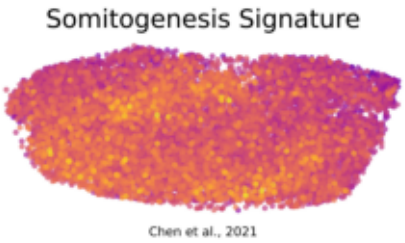
Pseudotime



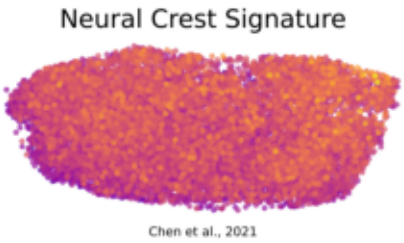
Gastrulation



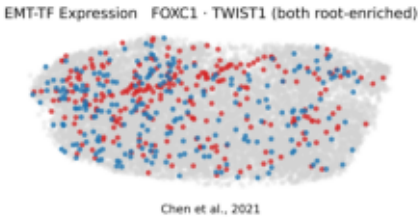
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Prostate adenocarcinoma (Chen et al., 2021), 10x — 14,266 malignant cells (12/13 samples above threshold).

Verdict.
Pattern reproduces as the non-branching case — one continuous gastrulation/somitogenesis → neural-crest developmental axis rooted at the Gast/pSom-high pole; no hub or bifurcation, so no cross-repression to test. Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. EMT-TFs FOXC1 / TWIST1 are both root-enriched — no expressed EMT-TF marks the neural-crest terminus.

Parameter choices

Min-cell threshold	55 (geom. mean, 125→25 gap; 12/13)	**Integration**	Harmony θ=10, n_pcs=20
Overdispersed genes	1,583	**Geometry**	Linear — non-branching curve
Root	Gast/pSom-high tip (origin)	**Module means**	Gast +0.004 / pSom +0.036 / Crest −0.069
Hub identity / r	NA (non-branching)	**EMT-TFs shown**	FOXC1 / TWIST1 (both root-enriched)